

Unit 8, Lesson 6: Solving Real-World Problems with Quadrilaterals



Benchmarks

MA.912.GR.1.4 Prove relationships and theorems about parallelograms. Solve mathematical and real-world problems involving postulates, relationships, and theorems of parallelograms.

MA.912.GR.1.5 Prove relationships and theorems about trapezoids. Solve mathematical and real-world problems involving postulates, relationships, and theorems of trapezoids.



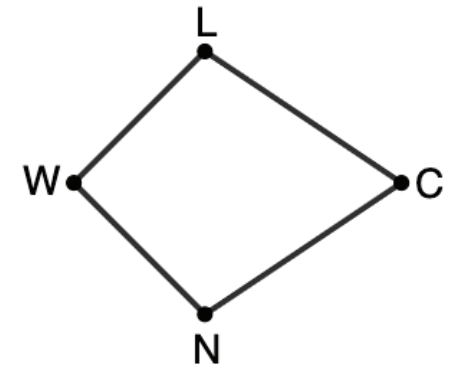
Learning Targets

- ☐ I can use properties of parallelograms to solve real-world problems.
- ☐ I can use properties of trapezoids to solve real-world problems.



8.6.1 Warm-Up: Error Analysis

Kite $WLCN$ is shown, with $m\angle W = 90^\circ$, $m\angle C = 68^\circ$, and $m\angle N = (6x + 5)^\circ$.



Marcia and Eugene both worked to determine the value of x . Their work is shown.

Marcia	Eugene
$90 + 90 + 68 + 6x + 5 = 360$ $6x + 5 + 248 = 360$ $6x + 253 = 360$ $6x = 107$ $x = \frac{107}{6}$	$360 - (90 + 68) = 202$ $6x + 5 = 202$ $6x = 197$ $x = \frac{197}{6}$

- Evaluate each student's work, then write a note to each student explaining the error they made.

Marcia	Eugene





8.6.2 Activity: Word Problem Gallery Walk

1. Work through the word problems with your partner. Show your work and write your answers in the appropriate boxes below.

Problem Number	Work	Answer
1		$KR =$
2		Area =
3		$m\angle MGS =$ $m\angle PSG =$ $m\angle GMP =$ $m\angle MPS =$

Problem Number	Work	Answer
4		$y =$
5		$FM =$ $RM =$ $TM =$
6		$DWFM$ _____ a parallelogram.

Problem Number	Work	Answer
7		The dimensions of $KRWF$ are
8		$DW =$



8.6.3 Your Turn!

- I am a quadrilateral with two pairs of opposite parallel sides. At least one of my angles has a measure of 90° and both of my diagonals measure 12 inches. Select all shapes that match this description.
 - ☐ Parallelogram
 - ☐ Rectangle
 - ☐ Square
 - ☐ Rhombus
 - ☐ Kite
- In your own words, write a description of a parallelogram that could **only** match a square.
- If a quadrilateral is a parallelogram, what additional information would be needed to show that the shape is a rectangle?
- If a quadrilateral is a parallelogram, what additional information would be needed to show that the shape is a rhombus?



8.6.4 Wrap-Up: Reflection

1. Circle the icon that best describes your understanding of each concept below.



I need help.



I got it.



I could help someone.

Properties of a parallelogram	  
Properties of a rectangle	  
Properties of a kite	  
Completing proofs using properties and theorems about parallelograms	  
Using the properties of a parallelogram to solve problems	  
Properties of a trapezoid	  
Using the properties of a trapezoid to solve problems	  

Unit 8, Lesson 7: Trapezoid Midsegment Theorem



Benchmark

MA.912.GR.1.5 Prove relationships and theorems about trapezoids. Solve mathematical and real-world problems involving postulates, relationships, and theorems of trapezoids.



Learning Targets

- ☐ I can prove the trapezoid midsegment theorem.
- ☐ I can use properties of trapezoids to solve mathematical and real-world problems.